

QUANTUM UNIVERSITY

MID SEMESTER EXAMINATION

Session 2023-24(Odd Semester) October 2023

ROLL NO.

--	--	--	--	--	--	--	--	--	--

Subject Code & Name: CS3402 & Computer Network

Program/Branch/Year: B.Tech 2nd CSE / SEM 3

Time: 2Hrs

Max Marks: 60

General Instructions. Read all instructions carefully.

1. Do not write anything on the question paper except your Roll No.
2. Answers should be written in clear and legible handwriting. Neatly labeled diagrams will fetch better marks.
3. Students must clearly write the question number & sub-part of question that they are attempting.
4. This question paper consists of 3 questions and all questions are compulsory. All questions carry 20 marks each.
5. Students are expected to take about 35 minutes each for Questions 1 to 3 and Remaining 15 minutes are for reading the question paper and revision

Parts i) and ii) of each question are compulsory and each part carries 2 marks. Parts iii), iv) and v) carry 8 marks each and the student may attempt any 2 parts.

Q1.

i) The first Network was called _____

- a) CNNET
- b) NSFNET
- c) ASAPNET
- d) ARPANET

(2 marks)

ii) The required resources for communication between end systems are reserved for the duration of the session between end systems in _____ method. (2 marks)

iii) Explain guided media differ from unguided media. Explain the three types of guided media and two types of unguided media. (8 marks)

iv) Differentiate circuit switching and packet switching? (8 marks)

v) Define different types of transmission mode and also its types. (8 marks)

Q2.

i) is a type of network interface. (2 marks)

ii) the layer of the OSI model is responsible for error-free data transmission (2 marks)

iii) Explain Data Link layer, and its primary function in the OSI model? (8 marks)

iv) Describe the difference between a MAC address and an IP address. How are they used in the Data Link layer and the Network layer, respectively? (8 marks)

v) Explain the term flow control, and how is it implemented in the Data Link layer? What are the advantages and disadvantages of different flow control methods? (8 marks)

Q3.

i) layer of the OSI model is responsible for routing data between different networks. (2 marks)

ii) Name the addressing scheme which is used by the IP protocol? (2 marks)

a) MAC addresses

b) IP addresses

c) URL addresses

d) DNS addresses

iii) Explain the role of the network layer in the OSI model? (8 marks)

iv) Name three network layer protocols used for routing data between networks and also explain in detail. (8 marks)

v) Compare and contrast IPv4 and IPv6, including their addressing schemes and any limitations or benefits. (8 marks)

12/10/23

Paishma

END SEMESTER EXAMINATION

Odd Semester 2023-2024

ROLL NO.

2	2	0	1	3	0	1	1	9	6
---	---	---	---	---	---	---	---	---	---

Subject Code & Name: CS3402 Computer Network

Program/Branch/Year: B.Tech CSE /SEM III DEC 2023

Time: 3 Hrs

Max Marks: 100

1. Do not write anything on the question paper except your Roll No.
2. Answers should be written in clear and legible handwriting. Neatly labeled diagrams will fetch better marks.
3. Students must clearly write the question number & sub-part of question that they are attempting.
4. This question paper consists of 5 questions and all are compulsory. All questions carry 20 marks each.
5. Students are expected to take about 30 minutes each for Questions 1 to 5 and remaining 30 minutes are for reading the question paper and revision.

Parts i) and ii) of each question are compulsory and each part carries 2 marks. Parts iii), iv) and v) carry 8 marks each and the student may attempt any 2 parts.

Q1.

i) _____ is an example of a peer-to-peer network

- a) Home network
- b) Corporate network
- c) Wide area network
- d) Metropolitan area network

(2 marks)

ii) _____ protocol is used for secure communication over the internet

- a) SSL/TLS
- b) FTP
- c) HTTP
- d) SMTP

(2 marks)

iii) Explain the TCP/IP model and its various layers in detail.

(8 marks)

iv) Discuss the metrics that influence the performance of computer network.

(8 marks)

Par

v) Describe network interfaces & its types in detail.

(8 marks)

Q2.

i) One of the following is not a function of the Data Link Layer

- a) Error detection and correction
- b) Framing
- c) Flow control
- d) Logical addressing

(2 marks)

ii) The purpose of a checksum in the Data Link Layer is _____

- a) To detect errors in the data
- b) To identify the source and destination of the data
- c) To regulate the flow of data between devices
- d) To provide security for the data

(2 marks)

iii) Discuss the various types of Data Link layer addressing, including unicast, multicast, and broadcast. List the benefits and drawbacks of each type.

(8 marks)

iv) Describe the process of error detection and correction in the Data Link layer. Discuss how different protocols handle errors.

(8 marks)

v) Explain TCP and UDP along with packet format.

(8 marks)

Q3.

i) _____ is not a valid IP address

- a) 192.168.0.1
- b) 10.0.0.256
- c) 172.16.0.0
- d) 255.255.255.0

(2 marks)

ii) _____ of the following is a private IP address

- a) 192.168.1.1
- b) 172.31.255.255
- c) 10.0.0.1
- d) All of the above

(2 marks)

iii) Compare and contrast IPv4 and IPv6, including their addressing schemes and any limitations or benefits.

(8 marks)

iv) Explain the different types of attacks that can be launched against the network layer, including denial-of-service (DoS) attacks and IP spoofing.

(8 marks)

v) Write short notes on:

- a) ICMP Protocol
- b) Net id
- c) Host id

(8 marks)

Q4.

i) _____ is considered to be more reliable,

- a) TCP
- b) UDP
- c) both a & b
- d) none of the above

(2 marks)

ii) The purpose of the Transmission Control Protocol (TCP) in the Transport layer is _____

- a) To provide error-free delivery of data
- b) To provide fast delivery of data
- c) To provide connectionless communication
- d) To provide congestion control

(2 marks)

iii) Explain the difference between connection-oriented and connectionless communication in the Transport layer, and give an example of a protocol that uses each type of communication.

(8 marks)

iv) Describe the three-way handshake process used by TCP for transferring the data.

(8 marks)

v) Explain the concept of flow control in the Transport layer, and describe how it is achieved in both TCP and UDP.

(8 marks)

Q5.

i) _____ Application layer protocol is used for real-time communication, such as video conferencing.

- a) SIP
- b) SNMP
- c) RTP
- d) SSH

(2 marks)

ii) _____ layer of the OSI model is responsible for providing services to end-users.

- a) Physical layer
- b) Data Link layer
- c) Transport layer
- d) Application layer

(2 marks)

iii) Describe the role of a DNS on a computer network with reference to its components.

(8 marks)

iv) Differentiate between HTTP, FTP and SMTP.

(8 marks)

v) Explain the importance of cryptography. Discuss two types of cryptography in detail.

(8 marks)